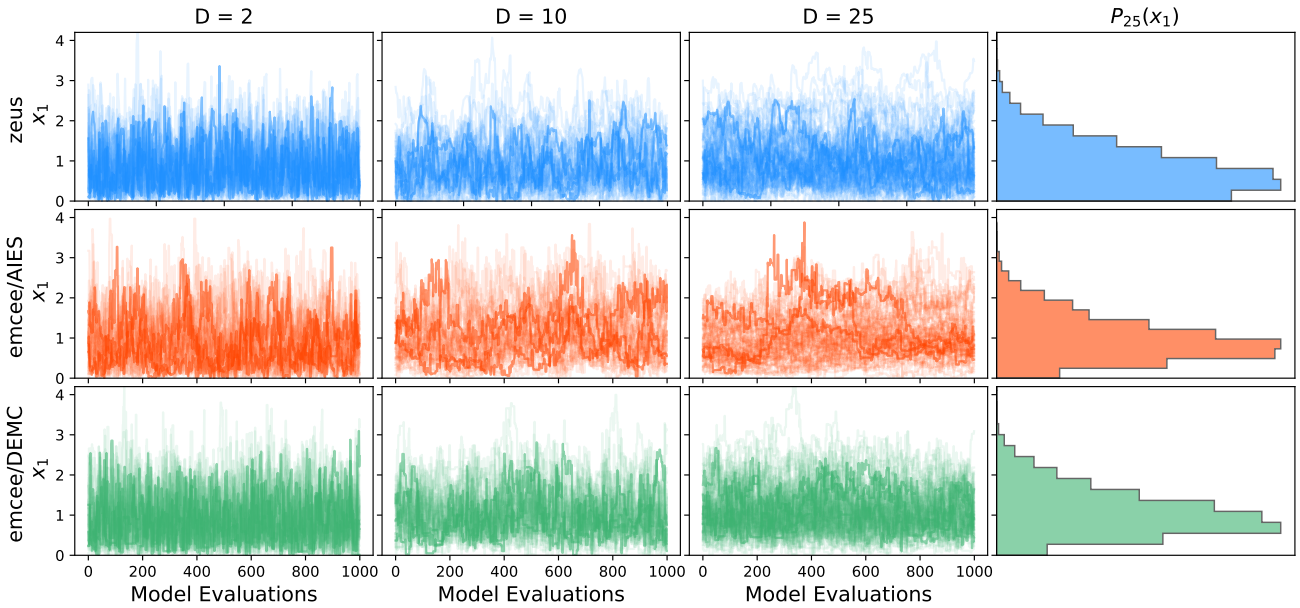

Figure 14. The figure shows numerical results (i.e. walker trajectories/chains for the first parameter) demonstrating the performance of the three ensemble MCMC methods in the case of the truncated normal distribution in 2, 10 and 25 dimensions respectively. The last column illustrates the 1-D marginal posterior corresponding to the first parameter x_1 estimated directly from the samples for the 25-dimensional case. `zeus` exhibits the least amount of bias near the hard boundary at zero compared to `emcee`/AIES and `emcee`/DEMC.

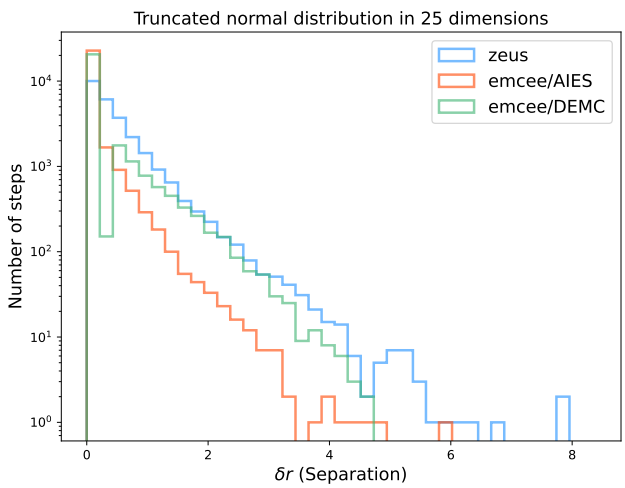

Figure 15. This figure shows the distribution of step sizes of walkers for the three different samplers in the case of the truncated normal distribution in $D = 25$. `zeus` and `emcee`/AIES exhibit similar distribution whereas `emcee`/DEMC performs shorter steps.

where $\mathbf{\Sigma}$ is the $p \times p$ positive semi-definite shape matrix and $\boldsymbol{\mu}$ is the mean vector.

We sampled the above distribution using the three samplers in 2, 10 and 25 dimensions respectively as shown in Figure 12. The diagonal elements of shape matrix $\mathbf{\Sigma}$ were set to 1 and the off-diagonal elements to 0.95. The mean vector $\boldsymbol{\mu}$ was set to $\mathbf{0}$. All three samplers managed to sample efficiently in 2, 10 and 25 dimensions as shown in Figure 12 and Table 1. Overall, `zeus` was the most efficient method with `emcee`/AIES being second and `emcee`/DEMC last. One can see from Figure 13 that the distributions of steps of `zeus` and `emcee`/AIES are very similar whereas that of `emcee`/DEMC is substantially shorter. Unlike the previous toy examples in which the proposal strategy of `emcee`/AIES was causing it to overshoot the bulk of posterior mass, in the case of the heavy-tailed t-distribution more proposals are accepted. On the other hand, `emcee`/DEMC's proposals which are optimised for Gaussian targets are more conservative in the case of the t-distribution and they do not extend far away. As also demonstrated in the previous toy examples, the locally adaptive nature of `zeus` allows it to perform efficient proposals that span large distances in parameter space.

3.1.5 The truncated normal distribution

The fifth and final toy example tests the case in which the target distribution is bounded from below or above. We chose to employ a truncated normal distribution similar to the one used in the first toy example, with the additional constraint being that $x > 0$. This effectively introduces a hard boundary along all dimensions. One of the reasons that we study this distribution is to assess the bias introduced by the presence of the hard boundary.

We sampled the above distribution using the three samplers in 2, 10 and 25 dimensions respectively as shown in Figure 14. The diagonal elements of the covariance matrix were set to 1 and the off-diagonal to 0.95. The mean vector $\boldsymbol{\mu}$ was set to $\mathbf{0}$. All three samplers managed to sample efficiently in 2, 10 and 25 dimensions as shown in Figure 14 and Table 1. Overall, `zeus` was the most efficient method with `emcee`/DEMC being second and `emcee`/AIES last. One can see from Figure 15 that the distributions of steps of `zeus` and `emcee`/AIES are very similar whereas that of `emcee`/AIES is slightly shorter. As shown in the right panels of Figure 14 `zeus` exhibits the least amount of bias compared to `emcee`/AIES and `emcee`/DEMC. In practical astronomical examples however, only one or two parameters would usually be bounded (e.g. the neutrino mass in galaxy clustering analyses) and thus unbiased sampling would be easier to perform by either of the three samplers.

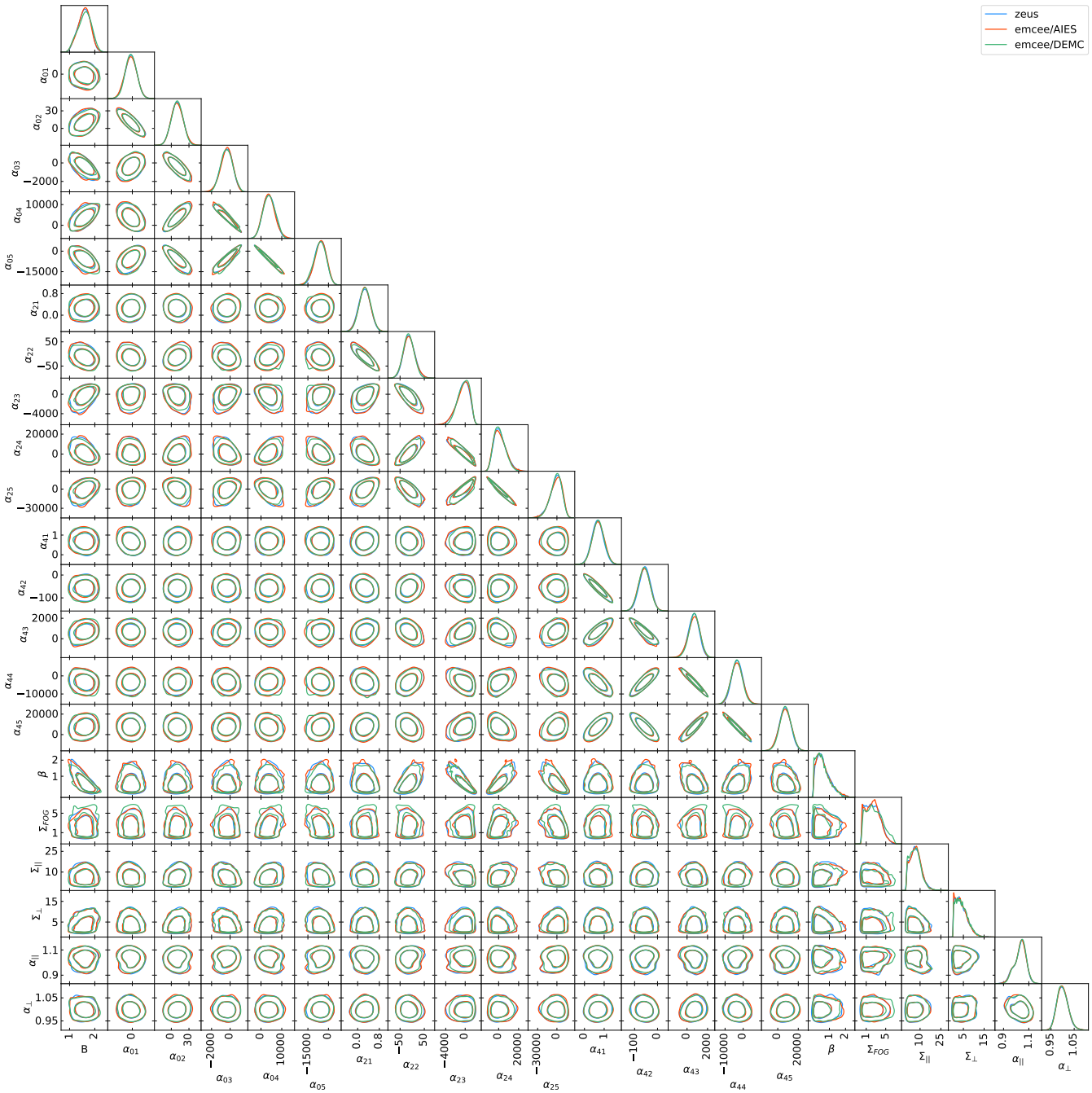

Figure 16. A corner plot showing the 1-D and 2-D marginalised posteriors for the 22-parameter Baryon Acoustic Oscillation model as produced by the three different ensemble MCMC methods.

3.2 Real astronomical analyses

The previous section employs toy examples in order to exhibit various scenarios that might emerge during sampling, and shows how `zeus` is better equipped to handle them. To demonstrate the efficiency of `zeus` compared to other samplers in realistic target distributions, we chose two common astronomical inference problems as the testing ground. Those are the cases of baryon acoustic oscillation (BAO) parameter inference and exoplanet parameter estimation.

We used the same three samplers in our comparison, namely `emcee` with AIES and DEMC, and of course `zeus`. We performed three distinct tests:

- The first test was to estimate the efficiency for each sampler, defined as the number of independent samples produced per log-likelihood evaluation. To this end, we ran the MCMC procedure 5 times for each sampler and computed the mean efficiency using the estimated autocorrelation time of the chains. The autocorrelation time was estimated using the method presented in Karamanis & Beutler (2020).

- The second test relates to the convergence rate of the three algorithms. As a measure of convergence rate, we adopt the inverse of the number of iterations required until all the convergence criteria specified below are met. In order to estimate the mean convergence rate

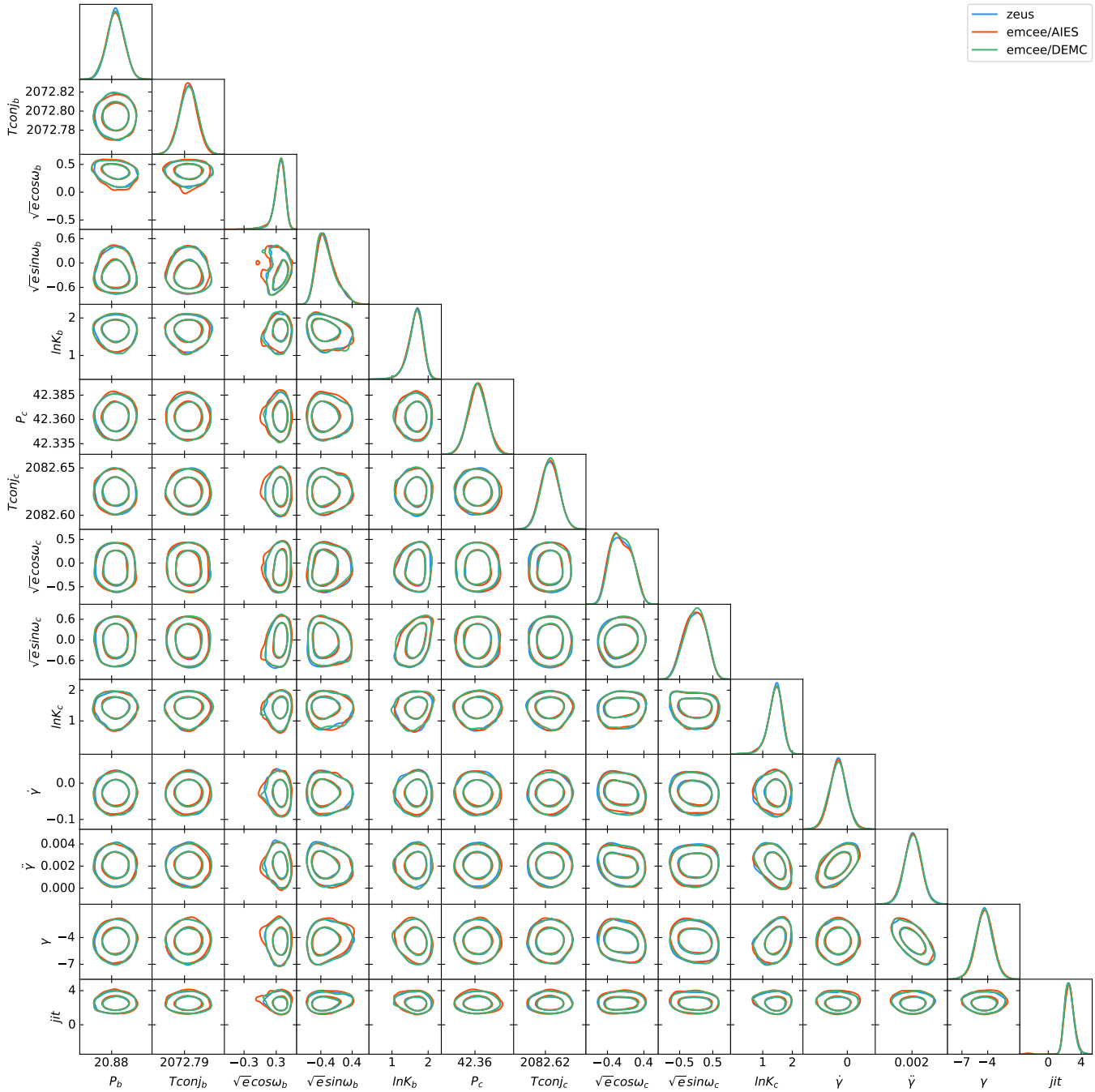

Figure 17. A corner plot showing the 1-D and 2-D marginalised posteriors for the 14-parameter radial velocity model as produced by the three different ensemble MCMC methods.

we ran the sampling procedure 40 times for each sampler initialising the walkers close to the *Maximum a Posteriori* (MAP) estimate.

- Finally, we tested the sensitivity of the samplers to the initial conditions by running 40 realisations with the walkers initialised from a small sphere (of radius 10^{-4}) around a randomly chosen point in the prior volume, counting how many of those attempts led to converged chains before a predetermined number of likelihood evaluations.

To determine whether a chain has converged we used four different metrics: the Gelman-Rubin split-R statistic (Gelman et al. 1992, 2013) using four independent ensembles of walkers; the Geweke test (Geweke 1992); a minimum length of the chain as a multiple of the integrated autocorrelation time (IAT); as well as an upper bound on the rate of change of the IAT. Only the second half of the chains was used to evaluate the aforementioned criteria. The number of walkers used in both examples was close to the minimum value of $2 \times D$ as specified below. As we will discuss in Section 4 this often leads to faster convergence.